Cope [3,3] | orbital phases along the reaction

REACTANT ORBITALS

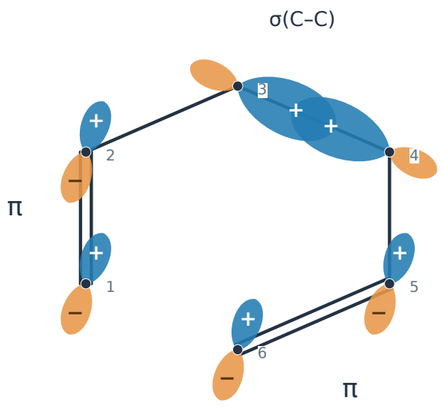
PROPOSED TRANSITION STATE

PRODUCT

THERMALLY ALLOWED

suprafacial / suprafacial

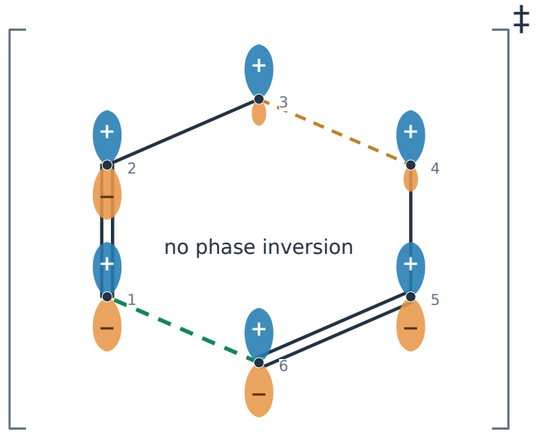
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

Δ

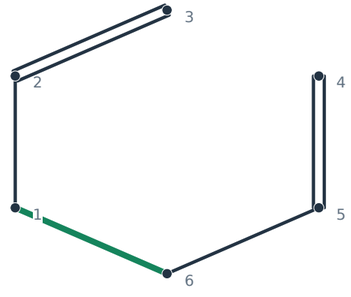
Hückel array · 6 electrons



Local overlap basis; σ centers reorganize.

Δ

Product connectivity



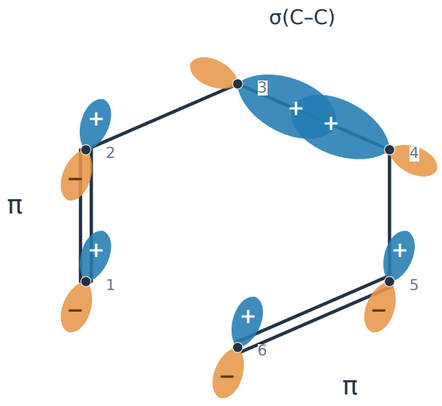
Rearranged hexa-1,5-diene

New σ bond(s) highlighted in green.

THERMALLY FORBIDDEN

suprafacial / antarafacial

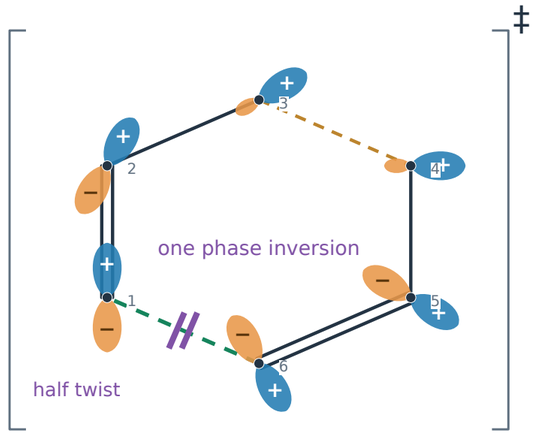
Occupied π + σ + π bond orbitals



Localized occupied orbitals; not one HOMO.

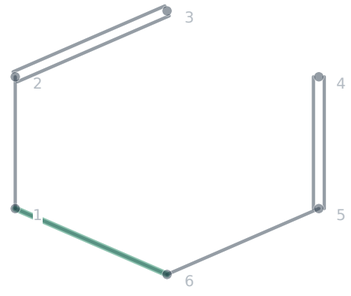
Δ

Möbius array · 6 electrons



Local overlap basis; σ centers reorganize.

Proposed product connectivity



Rearranged hexa-1,5-diene

Forbidden for this concerted thermal route.

+ / − orbital phase (not charge)

forming overlap / bond

phase mismatch

π/σ and transition-state local-basis drawings use different representations; neither is a calculated molecular HOMO.

Thermal, concerted pathways. Colored lobes are qualitative phase schematics; the middle panels are not optimized TS geometries or TS eigenorbitals.